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1. Which of the following is the identity matrix for 2x2 matrices?

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

Nice work

Correct. The identity matrix for 2x2 matrices is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

2. What is the determinant of the matrix $\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$?

12

18

14

7

Nice work

Correct. The determinant is calculated as $(3 \cdot 4) - (2 \cdot 1) = 10$.

3. Which of the following matrices represents a reflection in the x-axis?

$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Nice work

Correct. This matrix represents a reflection in the x-axis.

4. How is the inverse of a 2x2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ calculated, provided that the determinant is non-zero?

$\frac{1}{\det(A)} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$\frac{1}{\det(A)} \begin{pmatrix} a & -c \\ -b & d \end{pmatrix}$

$\frac{1}{\det(A)} \begin{pmatrix} d & -b \\ c & a \end{pmatrix}$

Nice work

Correct. This is the formula for the inverse of a 2x2 matrix.

5. What is the result of rotating a point (3,0) by 90 degrees counterclockwise using the rotation matrix?

(1,0)

(0,1)

(0,-1)

(1,1)

Nice work

Correct. Rotating (3,0) by 90 degrees counterclockwise gives (0,3).

6. Which transformation does the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ represent?

Rotation

Enlargement

Reflection

Shearing

Nice work

Correct. This matrix represents an enlargement by a factor of 2 in the x-direction and 3 in the y-direction.

7. If a matrix has a determinant of zero, what can be said about its inverse?

The inverse is negative.

The inverse is the same as the original matrix.

The inverse is the identity matrix.

The inverse does not exist.

Nice work

Correct. If the determinant is zero, the matrix is singular and does not have an inverse.

8. What happens to a shape when it is reflected in the line $y = x$ using the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$?

The shape is rotated by 90 degrees.

The x-coordinates are negated.

The x and y coordinates are swapped.

The shape is enlarged by a factor of 2.

Nice work

Correct. Reflecting in the line $y = x$ swaps the x and y coordinates.

9. Which matrix represents a shearing transformation along the x-axis?

$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Nice work

Correct. This matrix represents a shearing transformation along the x-axis.

10. Which of the following matrices represents a rotation by 180 degrees?

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Nice work

Correct. This matrix represents a rotation by 180 degrees.

11. Calculate the inverse of the matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$.

14/5 3/2

1 1/5 2/3

Feedback

The inverse of the matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ is $A^{-1} = \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$
1. Calculate the determinant of A:
 $\det(A) = (2 \cdot 4) - (3 \cdot 1) = 8 - 3 = 5$
2. Swap the numbers on the main diagonal and change the signs of the off-diagonal elements:
 $\text{adj}(A) = \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$
3. Divide each element of the adjugate matrix by the determinant:
 $A^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$

12. Find the determinant of the matrix $B = \begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$.

10/17 2/17

10/17 3/17

Feedback

The determinant of the matrix $B = \begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$ is 17.
1. Calculate the determinant:
 $\det(B) = (5 \cdot 3) - (2 \cdot -1) = 15 + 2 = 17$

13. Apply the transformation matrix $T = \begin{pmatrix} 4 & 2 \\ 0 & 1 \end{pmatrix}$ to the point (1, 1).

3 1

Feedback

The transformed point is (3, 1).
Solution:
1. Represent the point as a column vector:
 $P = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
2. Multiply the transformation matrix by the point vector:
 $TP = \begin{pmatrix} 4 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} (4 \cdot 1) + (2 \cdot 1) \\ (0 \cdot 1) + (1 \cdot 1) \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$

14. Determine the matrix representing a reflection in the line $y = -x$.

1/9 -1

1 4/9

Feedback

The matrix representing a reflection in the line $y = -x$ is $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.
1. The reflection matrix for the line $y = -x$ swaps and negates the coordinates.
Reflection matrix = $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

15. Calculate the image of the triangle with vertices (0, 0), (1, 0) and (0, 1) under the transformation matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

0,0

2,0

0,3

Feedback

The vertices of the transformed triangle are (0, 0), (2, 0) and (0, 3).
1. Transform each vertex using the given matrix:
 $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
 $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$
 $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$